

THE TANG–ZHANG SCHATTEN NORM FORMULA FAILS BELOW THE HILBERTIAN EXPONENT

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ABSTRACT. For $m \geq 2$, let $c_p(m)$ be the all-dimensional best constant in

$$\left\| \sum_{k=1}^m A_k \right\|_p \leq c_p(m) \left\| \sum_{k=1}^m |A_k| \right\|_p.$$

Tang and Zhang conjectured an explicit formula for every finite $p > 1$. We prove that their formula fails for every $1 < p < 2$. For each such p , an explicit pair of real 2×2 rank-one matrices gives a strictly larger ratio. The proof is analytic and reduces to one concavity estimate. At $p = 3/2$ we additionally give a fixed algebraic pair whose comparison is certified by seven strict rational inequalities. A transverse perturbation analysis identifies $p = 2$ as the exact local phase transition: below two the conjectured equiangular family is a non-smooth saddle, whereas at and above two it is locally maximizing in that direction. On the positive side, we prove the conjectured sharp bound for every family of rank-at-most-one summands when $2 \leq p < \infty$, and classify all equality cases. We also prove the corresponding endpoint statement for $p = \infty$. Finally, for arbitrary complex matrices we establish the conjectured sharp constant in the case $m = 2$, $p = 4$.

1. INTRODUCTION

For $A \in \mathbb{M}_n(\mathbb{C})$, write

$$|A| = (A^* A)^{1/2}$$

and let $\|A\|_p = (\text{Tr } |A|^p)^{1/p}$ be the Schatten p -norm for $1 \leq p < \infty$; $\|\cdot\|_\infty$ denotes the operator norm. For fixed $m, n \geq 1$, first define

$$c_p^{\text{abs}}(m, n) = \sup_{(A_1, \dots, A_m) \neq 0} \frac{\left\| \sum_{k=1}^m A_k \right\|_p}{\left\| \sum_{k=1}^m |A_k| \right\|_p}, \quad A_k \in \mathbb{M}_n(\mathbb{C}). \quad (1)$$

We use the dimension-free notation

$$c_p(m) = \sup_{n \geq 1} c_p^{\text{abs}}(m, n). \quad (2)$$

The all-zero family is excluded; the denominator otherwise cannot vanish.

2020 *Mathematics Subject Classification*. Primary 15A60; Secondary 15A45, 47B10.

Key words and phrases. Schatten norm, matrix absolute value, sharp constant, rank-one matrix, counterexample, phase transition, trace inequality.

Tang and Zhang [2] proved

$$c_1(m) = 1, \quad c_2(m) = \sqrt{\frac{1 + \sqrt{m}}{2}}, \quad c_\infty(m) = \sqrt{m},$$

and proposed a formula for the remaining exponents. For finite $p > 1$, consider

$$g_{p,m}(x) := x^p - 2x - (m - 1) = 0. \quad (3)$$

On $[1, \infty)$, one has $g_{p,m}(1) = -m < 0$, $g_{p,m}(x) \rightarrow \infty$, and $g'_{p,m}$ is strictly increasing. Thus $g_{p,m}$ either increases throughout this interval or first decreases to its unique minimum and then increases. In either case it has exactly one root $x_{p,m} > 1$. Their conjectured value is

$$C_{p,m}^{\text{TZ}} = \frac{\sqrt{x_{p,m}(x_{p,m} + m - 1)}}{(x_{p,m}^p + m - 1)^{1/p}}. \quad (4)$$

The lower bound $c_p(m) \geq C_{p,m}^{\text{TZ}}$ is attained by a rank-one equiangular family.

The dimension parameter matters at fixed size. Writing $d = \min\{m, n\}$, Zhang [3] subsequently obtained

$$c_1^{\text{abs}}(m, n) = 1, \quad c_2^{\text{abs}}(m, n) = \sqrt{\frac{1 + \sqrt{d}}{2}}, \quad c_\infty^{\text{abs}}(m, n) = \sqrt{d},$$

together with $c_p^{\text{abs}}(m, n) \leq d^{1/2-1/(2p)}$ for $1 \leq p \leq \infty$. Bourin and Lee [1] also highlighted the question for Schatten exponents other than two. Neither result asserts the explicit formula (4) for general p .

Our first result shows that this lower bound is never sharp below the Hilbertian exponent. The counterexample is dimension two, real, and rank one.

Theorem 1.1 (Failure throughout the sub-Hilbertian range). *For every $1 < p < 2$,*

$$c_p^{\text{abs}}(2, 2) > C_{p,2}^{\text{TZ}}.$$

More explicitly, let $x = x_{p,2} > 1$ solve $x^p - 2x - 1 = 0$, and put

$$s = \frac{x - 1}{x + 1}, \quad \delta = \frac{1}{2}x^{-p/(2-p)}. \quad (5)$$

With

$$e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u = \begin{pmatrix} 1 - \delta \\ \sqrt{2\delta - \delta^2} \end{pmatrix}, \quad v = \begin{pmatrix} s \\ \sqrt{1 - s^2} \end{pmatrix},$$

the two real rank-one matrices $A_1 = ee^T$ and $A_2 = uv^T$ satisfy

$$\frac{\|A_1 + A_2\|_p}{\| |A_1| + |A_2| \|_p} > C_{p,2}^{\text{TZ}}.$$

For instance, $[4/3, 5/3]$ is an explicit closed rational interval of failure. More generally, every closed subinterval of $(1, 2)$ has this property. In fact, Theorem 1.1 gives the maximal possible open interval below the known valid exponent $p = 2$.

Theorem 1.2 (Exact counterexample). *For $m = n = 2$ and $p = 3/2$, there are real rank-one matrices A_1, A_2 such that*

$$\frac{\|A_1 + A_2\|_{3/2}}{\| |A_1| + |A_2| \|_{3/2}} > \frac{207}{200} > C_{3/2,2}^{\text{TZ}}.$$

Consequently, the Tang–Zhang conjecture is false.

The failure occurs inside the rank-one class, but precisely on the opposite side of the Hilbertian exponent from the natural positive result.

Theorem 1.3 (Sharp rank-one bound). *Let $m \geq 2$, $2 \leq p < \infty$, and let $A_1, \dots, A_m \in \mathbb{M}_n(\mathbb{C})$ have rank at most one and not all vanish. Then*

$$\left\| \sum_{k=1}^m A_k \right\|_p \leq C_{p,m}^{\text{TZ}} \left\| \sum_{k=1}^m |A_k| \right\|_p. \quad (6)$$

The constant is sharp in the dimension-free rank-one problem and is attained whenever $n \geq m$. Equality holds precisely as follows, up to common input and output unitaries, a common positive scale, and the harmless phase changes in rank-one factorizations:

$$A_k = r u v_k^*, \quad \langle v_j, v_k \rangle = \begin{cases} 1, & j = k, \\ s_{p,m}, & j \neq k, \end{cases}$$

where $r > 0$, u is a unit vector, and

$$s_{p,m} = \frac{x_{p,m} - 1}{x_{p,m} + m - 1}.$$

In particular, equality requires $n \geq m$.

For $p = \infty$, the same rank-one argument gives the sharp constant $\sqrt{m}$, with the right vectors orthonormal. Our third result leaves the rank-one restriction entirely.

Theorem 1.4 (The full case $m = 2$, $p = 4$). *Let $A, B \in \mathbb{M}_n(\mathbb{C})$, and let $x > 1$ be the solution of*

$$x^4 - 2x - 1 = 0.$$

Then

$$\|A + B\|_4^4 \leq \frac{x^2(x+1)}{2} \| |A| + |B| \|_4^4. \quad (7)$$

The constant is sharp in the dimension-free problem, is attained for every $n \geq 2$, and equals $(C_{4,2}^{\text{TZ}})^4$.

The paper is organized as follows. Section 2 proves failure throughout $1 < p < 2$ and identifies the local transition at $p = 2$. Section 3 gives the fixed $p = 3/2$ counterexample and its rational certificate. Section 4 proves the sharp rank-one theorem and its equality statement. Section 5 proves Theorem 1.4.

2. FAILURE THROUGHOUT $1 < p < 2$

Proof of Theorem 1.1. Fix $1 < p < 2$, write $q = p/2 \in (1/2, 1)$, and use the parameters in (5). Since $x > 1$, one has $0 < s < 1$ and $0 < \delta < 1/2$, so the displayed vectors e, u, v are real unit vectors.

Let $U = (e, u)$ and $V = (e, v)$. The Gram matrices

$$L = U^T U = \begin{pmatrix} 1 & 1 - \delta \\ 1 - \delta & 1 \end{pmatrix}, \quad G = V^T V = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}$$

commute. Consequently, the squared singular values of $A_1 + A_2 = UV^T$ are

$$(2 - \delta)(1 + s), \quad \delta(1 - s),$$

whereas the eigenvalues of $|A_1| + |A_2| = VV^T$ are $1 + s$ and $1 - s$. Hence, if $R_p(\delta)$ denotes the attained ratio, then

$$R_p(\delta)^p = \frac{[(2 - \delta)(1 + s)]^q + [\delta(1 - s)]^q}{(1 + s)^p + (1 - s)^p}. \quad (8)$$

At $\delta = 0$, this is the Tang–Zhang value. Indeed,

$$1 + s = \frac{2x}{x + 1}, \quad 1 - s = \frac{2}{x + 1}, \quad x^p + 1 = 2(x + 1),$$

and direct substitution into (8) gives

$$R_p(0)^p = \frac{1}{2} x^q (x + 1)^{q-1} = (C_{p,2}^{\text{TZ}})^p. \quad (9)$$

It remains to compare the numerators in (8). Their difference is

$$\Delta_p(\delta) = (1 + s)^q ((2 - \delta)^q - 2^q) + (1 - s)^q \delta^q.$$

For $0 < \delta \leq 1$, the mean value theorem and $0 < q < 1$ give

$$2^q - (2 - \delta)^q \leq q\delta \leq \delta.$$

Since $(1 - s)/(1 + s) = x^{-1}$, it follows that

$$\begin{aligned} \Delta_p(\delta) &\geq (1 + s)^q (x^{-q} \delta^q - \delta) \\ &= (1 + s)^q \delta (x^{-q} \delta^{q-1} - 1). \end{aligned} \quad (10)$$

The choice $\delta = \frac{1}{2} x^{-q/(1-q)}$ in (5) satisfies

$$x^{-q} \delta^{q-1} = 2^{1-q} > 1.$$

Thus the right-hand side of (10) is strictly positive. Equations (8) and (9) now yield $R_p(\delta) > C_{p,2}^{\text{TZ}}$. $\square$

The preceding perturbation also explains why the proposed family ceases to be extremal exactly at $p = 2$. The phenomenon is not a regular Hessian eigenvalue crossing below two: a fractional-power cusp appears.

Proposition 2.1 (Local phase transition at $p = 2$). *Fix $p > 1$, put $q = p/2$, and let $s = (x_{p,2} - 1)/(x_{p,2} + 1)$. For*

$$\begin{aligned} u_\theta &= (\cos \theta, \sin \theta)^T, & v &= (s, \sqrt{1 - s^2})^T, \\ A_1 &= ee^T, & A_2(\theta) &= u_\theta v^T, \end{aligned}$$

let $N_p(\theta) = \|A_1 + A_2(\theta)\|_p^p$. Then

$$N_p(\theta) - N_p(0) = 2^{-q}(1 - s)^q|\theta|^p - q2^{q-2}(1 + s)^q\theta^2 + o(|\theta|^p + \theta^2). \quad (11)$$

Consequently, the equiangular Tang-Zhang candidate is transversely locally minimizing when $1 < p < 2$, but transversely locally maximizing when $p \geq 2$. For $1 < p < 2$, it is a non-smooth saddle: it remains locally maximizing along the common-left equiangular direction, while its transverse second derivative diverges to $+\infty$. The critical exponent is exactly $p = 2$.

Proof. As above, the squared singular values are $(1 + \cos \theta)(1 + s)$ and $(1 - \cos \theta)(1 - s)$. Therefore

$$N_p(\theta) = (1 + s)^q(1 + \cos \theta)^q + (1 - s)^q(1 - \cos \theta)^q.$$

The expansions

$$(1 + \cos \theta)^q = 2^q - q2^{q-2}\theta^2 + O(\theta^4)$$

and

$$(1 - \cos \theta)^q = 2^{-q}|\theta|^{2q} + O(|\theta|^{2q+2})$$

give (11). If $p < 2$, the positive $|\theta|^p$ term dominates the quadratic term; if $p > 2$, the negative quadratic term dominates. For $p = 2$, the exact difference is

$$N_2(\theta) - N_2(0) = -2s(1 - \cos \theta) < 0 \quad (\theta \neq 0 \text{ sufficiently small}).$$

Moreover, for $1 < p < 2$, differentiating the leading cusp twice away from zero shows

$$N_p''(\theta) \sim p(p-1)2^{-q}(1-s)^q|\theta|^{p-2} \longrightarrow +\infty.$$

Finally, with a common left vector and $y = (1 + s)/(1 - s)$, the ratio is

$$F_{p,2}(y) = \frac{\sqrt{y(y+1)}}{(y^p + 1)^{1/p}}.$$

Its logarithmic derivative has the sign of $2y + 1 - y^p$, which changes from positive to negative at the unique root $y = x_{p,2}$. Thus the same candidate is locally maximizing in the equiangular direction, completing the saddle and phase-transition assertions. $\square$

3. AN EXACT 2×2 COUNTEREXAMPLE

Set

$$e = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u = \begin{pmatrix} 39/40 \\ \sqrt{79}/40 \end{pmatrix}, \quad v = \begin{pmatrix} 5/8 \\ \sqrt{39}/8 \end{pmatrix}.$$

These are real unit vectors. Define

$$A_1 = ee^T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = uv^T = \begin{pmatrix} 39/64 & 39\sqrt{39}/320 \\ \sqrt{79}/64 & \sqrt{3081}/320 \end{pmatrix}. \quad (12)$$

Both matrices have rank one and unique nonzero singular value equal to one. Consequently,

$$|A_1| = ee^T, \quad |A_2| = vv^T.$$

Proof of Theorem 1.2. Let $U = (e, u)$ and $V = (e, v)$. Then $A_1 + A_2 = UV^T$, and the two Gram matrices are

$$L = U^T U = \begin{pmatrix} 1 & 39/40 \\ 39/40 & 1 \end{pmatrix}, \quad G = V^T V = \begin{pmatrix} 1 & 5/8 \\ 5/8 & 1 \end{pmatrix}.$$

The squared singular values of UV^T are the eigenvalues of LG . The two Gram matrices are simultaneously diagonalized by $(1, 1)^T$ and $(1, -1)^T$, so those squared singular values are

$$(1 + 39/40)(1 + 5/8) = \frac{1027}{320}, \quad (1 - 39/40)(1 - 5/8) = \frac{3}{320}. \quad (13)$$

On the other hand, the eigenvalues of $|A_1| + |A_2| = VV^T$ are

$$\frac{13}{8}, \quad \frac{3}{8}. \quad (14)$$

Thus, if

$$R = \frac{\|A_1 + A_2\|_{3/2}}{\||A_1| + |A_2|\|_{3/2}},$$

then

$$R^{3/2} = \frac{(1027/320)^{3/4} + (3/320)^{3/4}}{(13/8)^{3/2} + (3/8)^{3/2}}. \quad (15)$$

We first prove $R > 207/200$ using rational arithmetic only. Positivity allows us to raise each proposed enclosure to the fourth or second power. The exact differences are

$$\left(\frac{1027}{320}\right)^3 - \left(\frac{11989}{5000}\right)^4 = \frac{29144230879351}{4000000000000000} > 0, \quad (16)$$

$$\left(\frac{3}{320}\right)^3 - \left(\frac{301}{10000}\right)^4 = \frac{124819571}{4000000000000000} > 0, \quad (17)$$

$$\left(\frac{4143}{2000}\right)^2 - \left(\frac{13}{8}\right)^3 = \frac{773}{8000000} > 0, \quad (18)$$

$$\left(\frac{2297}{10000}\right)^2 - \left(\frac{3}{8}\right)^3 = \frac{5543}{200000000} > 0. \quad (19)$$

Put

$$A_0 = \frac{11989}{5000} + \frac{301}{10000} = \frac{24279}{10000}, \quad B_0 = \frac{4143}{2000} + \frac{2297}{10000} = \frac{5753}{2500},$$

and $q = 207/200$. A final exact comparison gives

$$A_0^2 - q^3 B_0^2 = \frac{1172956601313}{50000000000000} > 0. \quad (20)$$

Equations (16)–(20) imply

$$R^{3/2} > \frac{A_0}{B_0} > q^{3/2},$$

and hence $R > q$.

It remains to put the conjectured constant below the same rational separator. Let $x > 1$ solve $x^{3/2} - 2x - 1 = 0$, and write $x = t^2$. The unique positive root t of

$$h(t) = t^3 - 2t^2 - 1 = 0 \quad (21)$$

lies above $4/3$, where h is strictly increasing. Moreover,

$$h\left(\frac{1103}{500}\right) = \frac{310727}{125000000} > 0, \quad (22)$$

so $t < T := 1103/500$.

Writing $C = C_{3/2,2}^{\text{TZ}}$ and using $t^3 = 2t^2 + 1$, we obtain

$$C^6 = \frac{t^6}{16(t^2 + 1)} =: H(t).$$

The function H is strictly increasing for $t > 0$, since

$$H'(t) = \frac{t^5(2t^2 + 3)}{8(t^2 + 1)^2} > 0.$$

The remaining exact difference is

$$q^6 - H(T) = \frac{26731151399029597}{18772595200000000000} > 0. \quad (23)$$

It follows that $C^6 < H(T) < q^6$, hence $C < q < R$. $\square$

Remark 3.1. Numerically,

$$R = 1.0364136587048904\dots, \quad C_{3/2,2}^{\text{TZ}} = 1.0346539518514341\dots$$

These decimals play no role in the proof.

4. THE SHARP RANK-ONE PROBLEM FOR $p \geq 2$

We now prove Theorem 1.3. The proof reduces the matrix problem to a single scalar variable.

Proof of Theorem 1.3. Write

$$A_k = r_k u_k v_k^*, \quad r_k \geq 0,$$

where u_k, v_k are unit vectors whenever $r_k > 0$. Form the column matrices

$$U = (\sqrt{r_1}u_1, \dots, \sqrt{r_m}u_m), \quad V = (\sqrt{r_1}v_1, \dots, \sqrt{r_m}v_m)$$

and their Gram matrices

$$L = U^*U, \quad G = V^*V.$$

Then

$$\sum_k A_k = UV^*, \quad \sum_k |A_k| = VV^*, \quad (24)$$

and

$$\text{diag } L = \text{diag } G = (r_1, \dots, r_m), \quad \text{Tr } L = \text{Tr } G =: R > 0. \quad (25)$$

Let

$$K = L^{1/2}GL^{1/2}.$$

The nonzero eigenvalues of K are the squared singular values of UV^* ; the nonzero eigenvalues of G are the eigenvalues of VV^* . Therefore

$$\left\| \sum_k A_k \right\|_p^p = \text{Tr } K^{p/2}, \quad \left\| \sum_k |A_k| \right\|_p^p = \text{Tr } G^p. \quad (26)$$

Since $p/2 \geq 1$, the nonnegative eigenvalues of K give

$$\text{Tr } K^{p/2} \leq (\text{Tr } K)^{p/2}. \quad (27)$$

If $\lambda = \lambda_{\max}(G)$, then

$$\text{Tr } K = \text{Tr}(LG) \leq \lambda \text{Tr } L = R\lambda. \quad (28)$$

List the m eigenvalues of G , including zeros, as $\lambda = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 0$. Convexity gives

$$\text{Tr } G^p \geq \lambda^p + \frac{(R - \lambda)^p}{(m - 1)^{p-1}}. \quad (29)$$

If $\lambda = R$, then (26)–(28) give a ratio at most one, which is strictly smaller than the desired sharp constant. Suppose henceforth that $\lambda < R$, and put

$$y = \frac{(m - 1)\lambda}{R - \lambda}.$$

Since $\lambda \geq R/m$, one has $y \geq 1$. Combining (26)–(29) yields

$$\frac{\left\| \sum_k A_k \right\|_p}{\left\| \sum_k |A_k| \right\|_p} \leq F_{p,m}(y) := \frac{\sqrt{y(y + m - 1)}}{(y^p + m - 1)^{1/p}}. \quad (30)$$

Direct differentiation gives

$$\frac{d}{dy} \log F_{p,m}(y) = \frac{(m - 1)(2y + m - 1 - y^p)}{2y(y + m - 1)(y^p + m - 1)}.$$

For $p \geq 2$, the function $y^p - 2y - (m - 1)$ is strictly increasing on $[1, \infty)$, apart from an inessential zero derivative at the left endpoint when $p = 2$. It

has exactly one zero $x_{p,m} > 1$. Thus $F_{p,m}$ has a unique maximum at $x_{p,m}$, and (6) follows from (4).

We next track equality. Put

$$\beta = \frac{R - \lambda}{m - 1}.$$

Equality in the scalar maximization and in (29) forces

$$\lambda = x_{p,m}\beta, \quad \text{spec}(G) = \{\lambda, \beta, \dots, \beta\}.$$

The top eigenspace is one-dimensional. Equality in (28) forces the range of L into this eigenspace. If w is its unit eigenvector, then

$$L = Rww^*, \quad G = \beta I + (\lambda - \beta)ww^*. \quad (31)$$

The common diagonal condition (25) now gives

$$R|w_k|^2 = \beta + (\lambda - \beta)|w_k|^2.$$

Because $R = \lambda + (m - 1)\beta$, it follows that

$$|w_k|^2 = \frac{1}{m}, \quad r_k = \frac{R}{m} \quad (1 \leq k \leq m).$$

After simultaneous phase changes in the factorizations of the A_k , we may take $w = m^{-1/2}(1, \dots, 1)^T$. Equation (31) then says that the u_k coincide and

$$\langle v_j, v_k \rangle = \frac{\lambda - \beta}{R} = \frac{x_{p,m} - 1}{x_{p,m} + m - 1} \quad (j \neq k).$$

Conversely, this family takes equality at every step. The displayed Gram matrix is positive definite, so equality requires $n \geq m$. $\square$

Corollary 4.1 (Rank-one endpoint). *If $A_1, \dots, A_m$ have rank at most one, then*

$$\left\| \sum_k A_k \right\|_\infty \leq \sqrt{m} \left\| \sum_k |A_k| \right\|_\infty.$$

The constant is sharp in the dimension-free sense. Equality is possible only when $n \geq m$, and then holds precisely for equal nonzero singular values, a common one-dimensional range, and pairwise orthogonal right vectors, modulo the same unitary and phase symmetries as above.

Proof. Use the notation in the preceding proof and set $\lambda = \lambda_{\max}(G)$. Then

$$\|UV^*\|_\infty \leq \|U\|_\infty \|V^*\|_\infty = \sqrt{\|L\|_\infty \lambda} \leq \sqrt{R\lambda}.$$

Since $R = \text{Tr } G \leq m\lambda$, while $\|VV^*\|_\infty = \lambda$, the asserted inequality follows. Equality requires L to have rank one and $G = \lambda I_m$. The common diagonal condition then gives exactly the stated configuration. $\square$

5. THE FULL $m = 2$, $p = 4$ PROBLEM

Proof of Theorem 1.4. Put $H = |A|$ and $K = |B|$. Extend the partial isometries in the polar decompositions to unitaries, and write $A = UH$, $B = VK$. With $W = U^*V$, unitary invariance reduces the numerator to

$$S = H + WK.$$

Set

$$a = \operatorname{Tr} H^4, \quad b = \operatorname{Tr} K^4, \quad d = \operatorname{Tr} H^2 K^2.$$

The Hilbert–Schmidt triangle inequality applied to

$$SS^* = H^2 + HKW^* + WKH + WK^2W^*$$

gives

$$\|S\|_4^4 \leq (\sqrt{a} + \sqrt{b} + 2\sqrt{d})^2. \quad (32)$$

Introduce

$$e = \operatorname{Tr} H^3 K, \quad f = \operatorname{Tr} H K^3, \quad g = \operatorname{Tr} H K H K.$$

Cyclically collecting all words in the noncommutative expansion gives

$$\operatorname{Tr}(H + K)^4 = a + b + 4(e + f) + 4d + 2g. \quad (33)$$

Two lower bounds are needed. First,

$$e + f - 2d = \operatorname{Tr}((H - K)H(H - K)K) = \|H^{1/2}(H - K)K^{1/2}\|_2^2 \geq 0. \quad (34)$$

Second, the three-factor Schatten Hölder inequality gives

$$\begin{aligned} \sqrt{d} &= \|HK\|_2 = \|H^{1/2}(H^{1/2}K^{1/2})K^{1/2}\|_2 \\ &\leq \|H^{1/2}\|_8 \|H^{1/2}K^{1/2}\|_4 \|K^{1/2}\|_8 = a^{1/8} g^{1/4} b^{1/8}. \end{aligned}$$

Thus, when $ab > 0$,

$$g \geq \frac{d^2}{\sqrt{ab}}. \quad (35)$$

If $ab = 0$, one of H, K vanishes and the theorem is immediate. Combining (33)–(35), we obtain

$$\operatorname{Tr}(H + K)^4 \geq a + b + 12d + \frac{2d^2}{\sqrt{ab}}. \quad (36)$$

Let $A_0 = \sqrt{a}$, $B_0 = \sqrt{b}$, and define

$$z = \frac{A_0 + B_0}{\sqrt{A_0 B_0}} \geq 2, \quad s = \sqrt{\frac{d}{A_0 B_0}} \in [0, 1].$$

The upper bound on s is the Hilbert–Schmidt Cauchy–Schwarz inequality $d \leq \sqrt{ab}$. From (32) and (36),

$$\frac{\|A + B\|_4^4}{\|A\|_4^4 + \|B\|_4^4} \leq \Phi(z, s) := \frac{(z + 2s)^2}{z^2 - 2 + 12s^2 + 2s^4}. \quad (37)$$

If $s \geq 1/2$, then

$$2(z^2 - 2 + 12s^2 + 2s^4) - (z + 2s)^2 = (z - 2s)^2 + 16s^2 + 4s^4 - 4 \geq 0,$$

so $\Phi(z, s) \leq 2$.

Suppose $0 \leq s \leq 1/2$. The sign of $\partial\Phi/\partial z$ is the sign of

$$s^4 + 6s^2 - zs - 1.$$

Since $z \geq 2$,

$$s^4 + 6s^2 - zs - 1 \leq s^4 + 6s^2 - 2s - 1 \leq 2s - 1 \leq 0;$$

the middle inequality follows from $s^3 + 6s - 4 \leq 1/8 + 3 - 4 < 0$. Consequently,

$$\Phi(z, s) \leq \Phi(2, s) = f(s) := \frac{2(1+s)^2}{1+6s^2+s^4}. \quad (38)$$

The sign of $f'(s)$ is the sign of

$$1 - 6s - 2s^3 - s^4.$$

Hence f has a unique maximizer $s_0 \in (0, 1/2)$, characterized by

$$s_0^4 + 2s_0^3 + 6s_0 - 1 = 0. \quad (39)$$

The first branch cannot dominate, because $f(1/6) > 2$.

Set $x = (1 + s_0)/(1 - s_0)$. A direct calculation gives

$$(x^4 - 2x - 1)(1 - s_0)^4 = 2(s_0^4 + 2s_0^3 + 6s_0 - 1) = 0,$$

and

$$f(s_0) = \frac{x^2(x+1)^2}{x^4+1} = \frac{x^2(x+1)}{2}, \quad (40)$$

where the last equality uses $x^4 + 1 = 2(x+1)$. Equations (37)–(40) prove (7).

For sharpness, choose unit vectors r_1, r_2 with $\langle r_1, r_2 \rangle = s_0$, choose a unit vector u , and set

$$A = ur_1^*, \quad B = ur_2^*.$$

Then

$$\|A + B\|_4^4 = 4(1 + s_0)^2$$

and

$$\| |A| + |B| \|_4^4 = (1 + s_0)^4 + (1 - s_0)^4 = 2(1 + 6s_0^2 + s_0^4).$$

Their ratio is $f(s_0)$, proving sharpness. $\square$

Remark 5.1. Orthogonal direct sums of the two-dimensional extremal block give higher-dimensional equality examples. The proof above does not attempt a complete classification of all equality cases for Theorem 1.4.

REPRODUCIBILITY AND DISCLOSURE

The proof for the full interval $1 < p < 2$ is analytic; a standard-library high-precision script checks the closed formulas at representative rational exponents. The fixed $p = 3/2$ certificate consists of the seven positive rational differences in (16)–(23) and uses exact arithmetic only. Both scripts accompany this manuscript. Low-dimensional numerical searches were used for exploration and adversarial testing only.

OpenAI Codex assisted with proof exploration, counterexample search, adversarial checking, and manuscript preparation. The submitting author is responsible for the correctness of every statement and for compliance with the target journal’s authorship and disclosure policies.

REFERENCES

- [1] Jean-Christophe Bourin and Eun-Young Lee. Triangle inequalities for the operator symmetric modulus. *Proceedings of the American Mathematical Society*, July 2026. Early View.
- [2] Quanyu Tang and Shu Zhang. Generalizing Lee’s conjecture on the sum of absolute values of matrices. *Linear Algebra and its Applications*, 731:196–204, February 2026.
- [3] Teng Zhang. Operator symmetric moduli and sharp triangle inequalities. *Journal of the London Mathematical Society*, 114(2):e70672, August 2026.

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